

One State, Many Maximal Futures

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Abstract

For the initial-value problem $\dot{x} = f(x)$, $x(0) = 0$ with f continuous, $f(0) = 0$ and $f > 0$ off the origin, the structure of the solution set is classical: it was characterised by Wallach [14] in 1948 and treated in general form by Binding [4] in 1979. This note takes one concrete member of that class, $f(x) = |x|^{2/3} + x^2$, and works it out completely.

The example has exactly one solution defined on all of $[0, \infty)$, namely $x \equiv 0$, and continuum-many solutions that admit no forward extension: for each waiting time $T \geq 0$ there is a solution that rests at the origin until T , departs, and reaches $+\infty$ at $T + 3\sqrt{2}\pi/4$. Two independent mechanisms produce the two counts. Failure of local Lipschitz uniqueness at the equilibrium permits a continuum of waiting-time departures, which is what makes the second count $\mathfrak{c}$; finite-time escape prevents every non-trivial branch from being global, which is what leaves the first count at 1. Neither mechanism alone separates them.

The distinction this illustrates — between uniqueness among globally defined solutions and uniqueness among solutions that merely cannot be extended — is not introduced here either. It is discussed for general relativity by Earman [7], Smeenk and Wüthrich [13] and Doboszewski [5], for Newtonian blow-up by Wilson [15], and within a broader formulation of indeterminism by Azhar and Namjoo [2]. What is offered here is a minimal worked example in which every quantity is available in closed form. **The example does not by itself establish that any physical theory is indeterministic.** What it shows is that a uniqueness verdict depends on which class of solutions is quantified over; Section 7 sets out the competing readings and adjudicates between none of them.

1 Introduction

Here is a question with a less obvious answer than it looks.

If an initial state admits exactly one solution that exists for all future times, does it follow that the state has a unique possible future history?

Not necessarily — not if histories are taken to be solutions that cannot be extended any further, rather than solutions defined at every future time. The two classes can differ, and where they do, the uniqueness verdict can differ with them. That is the whole content of this note, and none of it is new.

We use *history* throughout for a solution regarded as a candidate physical evolution. Nothing in Sections 2–5 depends on that identification; Section 7 is where it does work, and says so.

The mathematics is classical. For scalar autonomous equations the solution set through an equilibrium was characterised by Wallach [14] in 1948, and the general theory — the exact conditions for departure from an equilibrium, for escape to infinity in finite time, and for the structure of the solution set — is given by Binding [4]. The conceptual point is also familiar. In general relativity the maximal globally hyperbolic development of initial data is unique while extensions past a Cauchy horizon are not, and whether to restrict attention to the well-behaved class is a live question rather than a settled one [7, 13]. In Newtonian mechanics Wilson [15] argues that failures of global existence are descriptive gaps rather than failures of determinism, in explicit opposition to Earman and Norton. Azhar and Namjoo [2] argue that a uniqueness-based criterion misses the sense in which a terminating history is itself a failure of determination.

What follows is therefore expository. Its purpose is to put the distinction into a small explicit example and work it out in full: a single scalar equation, elementary integrals, a blow-up time in closed form, an explicit solution family. No general relativity, no N -body mechanics, no partial differential equations, no geometry.

2 The example

Throughout we study

$$\dot{x} = f(x), \quad x(0) = 0, \tag{1}$$

for a continuous $f: \mathbb{R} \rightarrow \mathbb{R}$.

Definition 1. A *solution* of (1) is a differentiable function $x: I \rightarrow \mathbb{R}$ on an interval $I \subseteq [0, \infty)$ with non-empty interior and $\min I = 0$, satisfying $\dot{x}(t) = f(x(t))$ throughout, with one-sided derivatives at any endpoint belonging to I .

We work forward from 0 until Appendix A. The standing hypotheses are

(H1) f is continuous, $f(0) = 0$, and $f(x) > 0$ for $x \neq 0$;

(H2) $\int_0^1 \frac{dx}{f(x)} < \infty$;

(H3) $\int_1^\infty \frac{dx}{f(x)} < \infty$.

The example is

$$f(x) = |x|^{2/3} + x^2.$$

Four properties matter, and all four are elementary.

Lemma 2. For $f(x) = |x|^{2/3} + x^2$:

(i) (H1) holds.

(ii) f is not locally Lipschitz at the origin, but is C^1 on $\mathbb{R} \setminus \{0\}$, so the failure sits at exactly one point.

(iii) The substitution $x = u^3$ gives $\frac{dx}{x^{2/3} + x^2} = \frac{3 du}{1 + u^4}$ for $x > 0$.

(iv) (H2) and (H3) both hold.

Proof. (i) $|x|^{2/3}$ is the composition of $x \mapsto |x|$ with $s \mapsto s^{2/3}$ on $[0, \infty)$, so f is continuous on $\mathbb{R}$; $f(0) = 0$; and $f(x) > 0$ for $x \neq 0$ since $|x|^{2/3} > 0$ and $x^2 \geq 0$.

(ii) $|f(x) - f(0)|/|x| = |x|^{-1/3} + |x| \rightarrow +\infty$ as $x \rightarrow 0$, so no Lipschitz constant works on any neighbourhood of the origin. Away from it, $f'(x) = \frac{2}{3} \operatorname{sgn}(x)|x|^{-1/3} + 2x$ is continuous.

(iii) For $x > 0$ put $x = u^3$, a C^1 increasing bijection of $(0, \infty)$. Then $dx = 3u^2 du$, $x^{2/3} = u^2$, $x^2 = u^6$, and

$$\frac{dx}{x^{2/3} + x^2} = \frac{3u^2 du}{u^2(1 + u^4)} = \frac{3 du}{1 + u^4}.$$

(iv) By (iii) the two integrals become $\int_0^1 3 du/(1 + u^4) \leq 3$ and $\int_1^\infty 3 du/(1 + u^4) \leq \int_1^\infty 3u^{-4} du = 1$. $\square$

The two halves of (iv) do different work: (H2) is what allows a solution to leave the equilibrium at all, and (H3) is what makes the departing solution reach $+\infty$ in finite time. Both facts are classical, and Section 6 says whose they are. Because the substitution rationalises the integrand completely, the total is available in closed form.

Proposition 3 (The blow-up time).

$$T_* := \int_0^\infty \frac{dx}{x^{2/3} + x^2} = 3 \int_0^\infty \frac{du}{1 + u^4} = 3 \cdot \frac{\pi/4}{\sin(\pi/4)} = \frac{3\sqrt{2}\pi}{4} \approx 3.3321622036.$$

Proof. The first equality is Lemma 2(iii); the second is the standard evaluation $\int_0^\infty du/(1 + u^n) = (\pi/n)/\sin(\pi/n)$ at $n = 4$; the third is arithmetic, since $\sin(\pi/4) = \sqrt{2}/2$. $\square$

The rationale for this f is exactly that and no more: the substitution $x = u^3$ rationalises the integrand completely, reducing the escape integral to the elementary $3 \int_0^\infty du/(1 + u^4)$ and giving the convenient closed form $T_* = 3\sqrt{2}\pi/4$. We claim no uniqueness or optimality for the choice. Other members of the class are in some respects

more economical — for $f(x) = \sqrt{|x|}(1 + |x|)$ one has $F(x) = 2 \arctan \sqrt{x}$, hence $T_* = \pi$ and the elementary family $x_T(t) = \tan^2((t - T)/2)$, whereas here φ itself has no elementary closed form. Nothing in what follows depends on the choice; any member of the class would serve.

3 The solution family

Define, for $x > 0$,

$$F(x) = \int_0^x \frac{du}{f(u)}. \quad (2)$$

Lemma 4. *Assume (H1)–(H3). Then F is a C^1 strictly increasing bijection from $(0, \infty)$ onto $(0, T_*)$ with $F(0^+) = 0$. Let $\varphi = F^{-1}$, extended by $\varphi(0) = 0$. Then φ is defined on $[0, T_*)$, is C^1 there including at 0, satisfies $\varphi' = f(\varphi)$ and $\varphi'(0^+) = 0$, and $\varphi(t) \rightarrow +\infty$ as $t \rightarrow T_*^-$.*

Proof. $F' = 1/f > 0$ is continuous on $(0, \infty)$, so F is C^1 and strictly increasing there; (H2) gives $F(0^+) = 0$ and (H3) gives $F(\infty^-) = T_* < \infty$, so F is onto $(0, T_*)$. Hence $\varphi = F^{-1}$ is defined on $(0, T_*)$, and $\varphi' = 1/F'(\varphi) = f(\varphi)$ there; $\varphi(t) \rightarrow +\infty$ as $t \rightarrow T_*^-$ because F is a bijection onto $(0, T_*)$.

It remains to treat $t = 0$, which is what lets the pieces glue in Proposition 5. Since $F(0^+) = 0$ and F is increasing, $\varphi(t) \rightarrow 0$ as $t \rightarrow 0^+$, so φ is continuous at 0 with $\varphi(0) = 0$. Then $\varphi'(t) = f(\varphi(t)) \rightarrow f(0) = 0$ by continuity of f . Now φ is continuous on $[0, t]$ and differentiable on $(0, t)$, so the mean value theorem gives $(\varphi(t) - \varphi(0))/t = \varphi'(\xi_t)$ for some $\xi_t \in (0, t)$; letting $t \rightarrow 0^+$ forces $\xi_t \rightarrow 0$ and hence $\varphi'(\xi_t) \rightarrow 0$. So the right derivative of φ at 0 exists and equals $0 = f(\varphi(0))$, and φ' is continuous at 0. $\square$

Proposition 5 (The waiting-time family). *For each $T \in [0, \infty)$ define*

$$x_T(t) = \begin{cases} 0, & 0 \leq t \leq T, \\ \varphi(t - T), & T < t < T + T_*, \end{cases}$$

and let $x_\infty \equiv 0$ on $[0, \infty)$. Then every x_T and x_∞ solves (1), and each x_T is C^1 on $[0, T + T_)$.*

Proof. On $[0, T)$ and on $(T, T + T_*)$ this holds by construction and Lemma 4. At $t = T$ the left derivative is 0, the right derivative is $\varphi'(0^+) = 0$ by Lemma 4, and $f(x_T(T)) = f(0) = 0$; so the two-sided derivative exists and equals $f(x_T(T))$. The join is therefore C^1 , not merely continuous. That x_∞ solves (1) is immediate from $f(0) = 0$. $\square$

In words: the solution rests at the equilibrium as long as it likes, leaves with zero slope at a time of its choosing, and then — after exactly $T_* = 3\sqrt{2}\pi/4$ more, the same for every branch — runs off to $+\infty$.

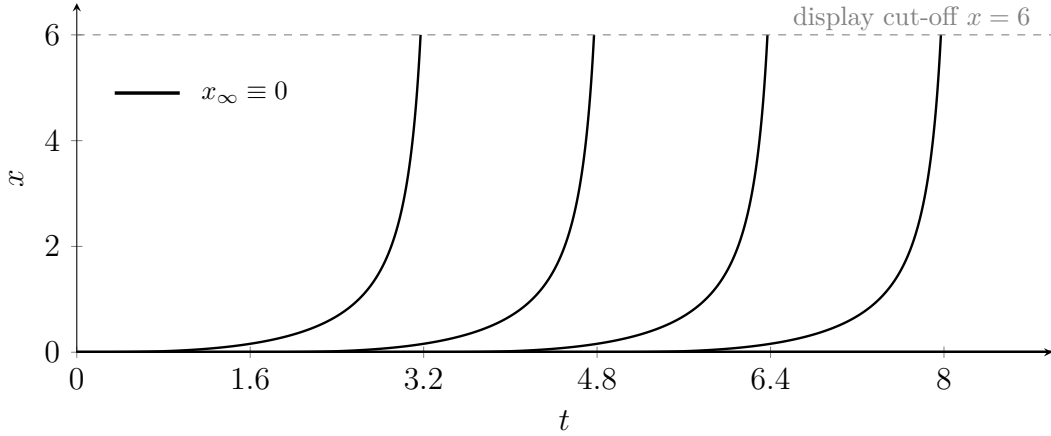


Figure 1: Four members of the family, departing at $T = 0, 1.6, 3.2, 4.8$. Each curve is φ , traced *for the drawing only* by sampling x and plotting $(T + F(x), x)$, so that F is evaluated and never inverted; the blow-up time is the analytic $T_* = 3\sqrt{2}\pi/4$ of Proposition 3, and nothing here is offered as evidence for it. The curves are drawn only up to $x = 6$; that ceiling is graphical, and each branch in fact runs to $+\infty$ as t approaches $T + T_*$. Along the axis is x_∞ , the one solution still defined at every later time.

4 Globally defined versus forward-maximal

Two classes of forward solution, kept apart.

Definition 6. $\mathcal{H}_{\text{glob}}^+(0)$ is the set of solutions of (1) defined at *every* $t \geq 0$, that is, those with $I = [0, \infty)$.

$\mathcal{H}_{\text{max}}^+(0)$ is the set of solutions of (1) on intervals $I \subseteq [0, \infty)$ admitting no extension to a solution on a strictly larger $J \subseteq [0, \infty)$. We call these *forward-maximal*; Section 5 explains why the prefix is not optional.

Plainly $\mathcal{H}_{\text{glob}}^+(0) \subseteq \mathcal{H}_{\text{max}}^+(0)$: a solution on $[0, \infty)$ has nowhere further to go inside $[0, \infty)$. The main statement is that the two classes are nevertheless as far apart as they could be.

Theorem 7. Assume (H1)–(H3); in particular $f^{-1}(0) = \{0\}$, which is part of (H1). Then, with $\mathcal{H}_{\text{max}}^+(0)$ the forward-maximal solutions of Definition 6 and not the two-sided maximal solutions of Section 5,

$$\mathcal{H}_{\text{glob}}^+(0) = \{x_\infty\}, \quad \mathcal{H}_{\text{max}}^+(0) = \{x_T : T \in [0, \infty)\} \cup \{x_\infty\},$$

so that $|\mathcal{H}_{\text{glob}}^+(0)| = 1$ while $|\mathcal{H}_{\text{max}}^+(0)| = \mathfrak{c}$.

Proof. We argue in eight steps.

Step 1 (monotonicity). Since $f \geq 0$ everywhere, every solution has $\dot{x} \geq 0$ and so is non-decreasing; as $x(0) = 0$, every solution is non-negative.

Step 2 (the zero set). Let x be a solution on I and put $Z = \{t \in I : x(t) = 0\}$. Then Z is closed in I by continuity, and is an interval: if $t_1 < t < t_2$ with $t_1, t_2 \in Z$ then $0 = x(t_1) \leq x(t) \leq x(t_2) = 0$ by Step 1. Also $0 \in Z$. Set $T = \sup Z \in [0, \infty]$.

Step 3 (each x_T is forward-maximal). By Lemma 4, $x_T(t) \rightarrow +\infty$ as $t \rightarrow (T + T_*)^-$. Any $J \subseteq [0, \infty)$ strictly containing $I = [0, T + T_*)$ and admitting a solution extending x_T would have to contain the point $T + T_*$, at which the extension would have to be finite and continuous — impossible. So $x_T \in \mathcal{H}_{\max}^+(0)$. Likewise $x_\infty \in \mathcal{H}_{\text{glob}}^+(0) \subseteq \mathcal{H}_{\max}^+(0)$.

Step 4 (no forward-maximal solution has a compact domain). Suppose $x \in \mathcal{H}_{\max}^+(0)$ has $I = [0, b]$ with $b < \infty$. Then $x(b)$ is finite and f is continuous, so Peano's existence theorem supplies a solution y of $\dot{y} = f(y)$ on some $[b, b + \delta]$ with $y(b) = x(b)$. Concatenating x and y gives a solution on $[0, b + \delta]$ properly extending x , contradicting maximality. Hence every I arising in $\mathcal{H}_{\max}^+(0)$ is right-open or is $[0, \infty)$.

Step 5 (the case $T = \infty$). If $T = \infty$ then Z is unbounded, and being an interval containing 0 it equals I ; so $x \equiv 0$ on I . If $I \neq [0, \infty)$ then extending by 0 gives a solution on a strictly larger subinterval of $[0, \infty)$, contradicting maximality. So $I = [0, \infty)$ and $x = x_\infty$.

Step 6 (the case $T < \infty$). First, $T \in I$. Otherwise I is an interval containing $[0, T)$ but not T , hence $I \subseteq [0, T)$ and $x \equiv 0$ on I . Then x has the finite right endpoint $T \notin I$ at which it remains finite — indeed $x(t) \rightarrow 0$ as $t \uparrow T$ — so setting $x(T) = 0$ extends it continuously, and the extension solves (1) at T because $f(0) = 0$. (In general this is the continuation principle used in Step 4: at a finite right endpoint where the solution stays finite, Peano's theorem supplies a local solution from the limiting value, so the solution extends. Here the limiting value is the equilibrium and the extension is explicit.) Either way maximality is contradicted.

For $t \in I$ with $t > T$ we have $x(t) \neq 0$ by the definition of T as $\sup Z$, and $x(t) \geq x(T) = 0$ by Step 1; hence $x(t) > 0$. On $I \cap (T, \infty)$ therefore $\dot{x} = f(x) > 0$, so x is strictly increasing there and $\dot{x}/f(x) = 1$. Integrating from s to t with $T < s < t$ and substituting $u = x(\sigma)$ — legitimate since x is a C^1 strictly increasing bijection from $[s, t]$ onto $[x(s), x(t)]$ — gives

$$t - s = \int_s^t \frac{\dot{x}(\sigma)}{f(x(\sigma))} d\sigma = \int_{x(s)}^{x(t)} \frac{du}{f(u)} = F(x(t)) - F(x(s)).$$

Letting $s \downarrow T$ and using $x(s) \rightarrow x(T) = 0$ together with $F(0^+) = 0$ yields $F(x(t)) = t - T$, that is $x(t) = \varphi(t - T)$. Since F takes values in $(0, T_*)$ this forces $t - T < T_*$, so $I \subseteq [0, T + T_*)$. By Step 4 the domain is not compact, and by maximality together with Step 3 it cannot be a proper subinterval of $[0, T + T_*)$; hence $I = [0, T + T_*)$ and $x = x_T$.

Step 7 (the cardinality of $\mathcal{H}_{\max}^+(0)$). Steps 3, 5 and 6 together give the stated description of $\mathcal{H}_{\max}^+(0)$. For the lower bound, $T \mapsto x_T$ is injective: if $T_1 < T_2$ then on the interval $(T_1, T_2]$, of positive length, $x_{T_1} = \varphi(\cdot - T_1) > 0$ while $x_{T_2} = 0$, so the two functions differ. Moreover $x_\infty \neq x_T$ for every $T < \infty$, since $x_\infty \equiv 0$ while $x_T > 0$ on $(T, T + T_*)$; so the listed solutions are pairwise distinct and $|\mathcal{H}_{\max}^+(0)| \geq |[0, \infty)| = \mathfrak{c}$.

For the upper bound, every solution is a continuous real-valued function on a subinterval of $[0, \infty)$. There are $\mathfrak{c}$ such subintervals, and a continuous function on an interval is determined by its values on a countable dense subset, so each interval carries at most $\mathfrak{c}^{\aleph_0} = \mathfrak{c}$ of them. Hence $|\mathcal{H}_{\max}^+(0)| \leq \mathfrak{c} \cdot \mathfrak{c} = \mathfrak{c}$. Combining the bounds, $|\mathcal{H}_{\max}^+(0)| = \mathfrak{c}$.

Step 8 (the globally defined solutions). Let $x \in \mathcal{H}_{\text{glob}}^+(0)$, so $I = [0, \infty)$. A solution on $[0, \infty)$ admits no extension inside $[0, \infty)$, so $\mathcal{H}_{\text{glob}}^+(0) \subseteq \mathcal{H}_{\max}^+(0)$ and x is on the list above. It is not any x_T with $T < \infty$: by Lemma 4 such a branch satisfies $x_T(t) \rightarrow +\infty$ as $t \uparrow T + T_*$, so it is defined only on the bounded interval $[0, T + T_*)$ and in particular not at $t = T + T_*$. Every non-zero branch therefore reaches $+\infty$ after the finite additional time T_* and fails to be global. The only remaining possibility is $x = x_\infty$, which is indeed defined on all of $[0, \infty)$. Hence $\mathcal{H}_{\text{glob}}^+(0) = \{x_\infty\}$ and $|\mathcal{H}_{\text{glob}}^+(0)| = 1$. $\square$

Remark 8 (The counts do not depend on the solution concept). Definition 1 asks for a differentiable solution; Binding [4] works instead with absolutely continuous x satisfying $\dot{x} = f(x)$ almost everywhere. The two give the same sets here. If x is absolutely continuous with $\dot{x} = f(x)$ a.e., then $x(t) = x(0) + \int_0^t f(x(s)) ds$, and $f \circ x$ is continuous, so x is C^1 and solves (1) in the classical sense. So Theorem 7 is not an artefact of the regularity demanded of solutions.

Remark 9 (Each hypothesis moves one count). Without (H2) no branch could leave the origin, and both counts would be 1; without (H3) every x_T would be defined on all of $[0, \infty)$, and both counts would be $\mathfrak{c}$. Only the conjunction separates them. The margin is also the largest available, since by Step 7 no solution set of (1) can exceed $\mathfrak{c}$.

5 Three senses of “maximal”

The word carries at least three meanings in the surrounding literature, and conflating them produces false statements about this example. We separate them once and then keep the prefix.

- **Forward-maximal:** no extension within $[0, \infty)$. These are the x_T , on $[0, T + T_*)$ — the class counted in Theorem 7.
- **Maximal (standard):** no extension to any larger interval of $\mathbb{R}$. *No x_T is maximal in this sense.* Setting $x(t) = 0$ for $t < 0$ always extends it, since $f(0) = 0$. Appendix A classifies the genuinely maximal solutions.
- **Maximal (extremal):** the pointwise *largest* solution. Here that is x_0 , which departs immediately; the pointwise smallest is x_∞ . Hartman [10, ch. III §2] uses the word in this sense.

So $[0, T + T_*)$ is a forward-maximal interval and not a maximal interval, and $\mathcal{H}_{\max}^+(0)$ is not a set of maximal solutions.

6 What is classical, and where

None of the mathematics above is new. The attributions below are specific rather than general, because a general gesture at “the classical theory” would leave a reader unable to check them.

Wallach [14], 1948. Characterised *all* solutions of $\dot{x} = f \circ x$, $x(0) = 0$ for continuous f . Binding reports this at [4, p. 184] — “Wallach characterized all solutions of (1.1)” — and credits the monotonicity lemma to [14, §2] and the structure theorem to [14, §7]. *We know this paper only through Binding; we have not read it.*

Binding [4, §3, pp. 186–188]. The machinery of Theorem 7. Lemma 3.2: every solution is monotonic, which Binding credits to Wallach [14, §2] for continuous f — our Step 1. Theorem 3.3: the separation identity $\lambda([s, t] \setminus \dot{x}^{-1}(0)) = \int_{x(s)}^{x(t)} 1/f$ in measure-theoretic form, of which the computation in Step 6 is the elementary case. Theorem 3.5: necessary conditions for local existence, reducing for continuous f with $x > 0$, in his words, to “ $f > 0$ and $1/f$ integrable, as given by Wallach”.

Binding [4, §§5–6, pp. 190–193]. Theorem 5.3 is the uniqueness criterion, in the Osgood [12] and Kamke line — Binding names both on the same page — and see Hartman [10, ch. III §6], where Theorem 6.1 is Kamke’s and Corollary 6.2 is Osgood’s. Fjordholm [8], whose own subject is Filippov uniqueness for discontinuous fields, restates the continuous case as “it was shown by Binding that the solution is unique if and only if b satisfies the so-called Osgood condition at all zeroes of b ”. Case (6.5) is exactly hypothesis (H3): when $\int^\infty 1/f < \infty$, Binding writes, “we have the phenomenon of escape to infinity for at least one solution”; his case (6.2), where it diverges, is the complementary one in which the branch survives on the whole half-line. Lemma 6.3 is a duality — $\hat{f}(\alpha) = \alpha^2 f(1/\alpha)$ turns escape to infinity into non-uniqueness at the origin — so the observation that (H2) and (H3) are the two ends of one integral is his; he credits awareness of it to Wintner and to Brauer and Sternberg.

Binding [4, Lemma 7.5, p. 194]. The structure theorem: every solution is obtained from the extremal one by inserting constant segments at zeros of f . With $f^{-1}(0) = \{0\}$ that is precisely the waiting-time family of Proposition 5. Binding notes that Wallach’s proof for continuous f carries over.

Hartman [10, ch. III §5 Thm. 5.1, p. 30]. A theorem of Wintner: if $\int^\infty du/\psi = \infty$ and $|f| \leq \psi$, solutions exist on the whole strip; finite-time escape requires the negation. Exercise 6.4 (p. 33), attributed in the chapter Notes to Wallach, states the departure criterion as an exact dichotomy; it is also the standard source for the textbook waiting-time examples $\dot{x} = x^{1/3}$ and $\dot{x} = \sqrt{|x|}$, whose continuum of wait-then-depart solutions is the classical Peano phenomenon and is not due to any recent author. The general fact that non-uniqueness forces continuum-many solutions — our Step 7, in far greater generality — is Kneser’s theorem, [10, ch. II §4 Thm. 4.1].

Drivas and Mailybaev [6] (preprint 2018; published 2021). Cited here for continuation past a non-Lipschitz singularity and the regularisation that selects among the continuations — the context their paper is actually about — and not as the origin of the waiting-time family, which is classical. In passing they record it for $\dot{x} = x^{1/3}$: “a

continuum infinity of solutions, which stay at the origin until an arbitrary time t_1 and peel [sic] off.” There the branches are global, so no gap between the two counts arises.

Berndl et al. [3], 1995. Use “maximal (non-extendible) solution” against “global existence” as basic vocabulary in a foundational setting, and list escape to infinity in finite time among the ways a solution may cease to exist.

Given Lemma 7.5 and case (6.5) together, Theorem 7 follows in two lines; the self-contained proof above is written out because it is short, not because it is new. The wider uniqueness and non-uniqueness literature is surveyed in Agarwal and Lakshmikantham [1].

On the particular choice of f : we have not found a prior publication using $|x|^{2/3} + x^2$ to illustrate this point, though the search behind that remark is bounded and we make no priority claim. What can be said without qualification is that the choice gives a compact closed-form blow-up time. It is not the most economical member of the class — Section 2 names one that is — and we claim no advantage for it beyond convenience.

7 Bearing on determinism

The example admits at least three readings. **None of them is introduced here.** We set them out with attributions and do not adjudicate between them.

First, what is being tested. We do not propose a definition of determinism; we borrow one already in use, so that the readings below have something definite to disagree about. Doboszewski [5, p. 2] adopts the criterion of Butterfield (1989) — his DM2 — under which a theory is deterministic for a given kind of region when any two models agreeing on a region of that kind agree throughout, so that, in his gloss, “to find a witness of indeterminism one needs merely to find two models that are isomorphic in region S but are not isomorphic overall — just as standard Laplacian intuitions demand.” Transposed to the present setting with all due looseness, the region is the initial state $x(0) = 0$, the models are the solutions through it, and the question is whether agreement there forces agreement throughout. **Everything then turns on what “throughout” ranges over** — which is precisely the choice Theorem 7 makes visible. Other criteria are in circulation and give different verdicts on this example; we take no stand on which is correct, and the example is not evidence for any of them.

Reading 1: the global-solution criterion

If a physical history must be defined at every time, the branches x_T are not histories, the state $x = 0$ has exactly one future, and the example says something about an initial-value problem and nothing about determinism. This is close to the position Wilson [15, p. 182] defends for Newtonian mechanics: failures of global existence are “descriptive holes”, and “it is as descriptive holes such issues should be addressed, *not as failures of determinism*.” He names Earman and Norton as holding the contrary view and calls their classification “terminologically unfortunate”.

Reading 2: the maximal-history criterion

If a solution that cannot be extended counts as a history even though it ends, the same state has continuum-many futures, and requiring totality looks like a stipulation that secures determinism rather than a discovery that it holds. The structure of this objection is familiar from general relativity, where the maximal globally hyperbolic development of initial data is unique but, in Earman’s phrase [7, §6.4], often “not maximal simpliciter”, and the larger extensions are not unique. Smeenk and Wüthrich [13] organise their treatment around whether global hyperbolicity should be imposed by fiat, established from weaker assumptions, or justified from outside the theory — their §5 is titled “Global Hyperbolicity by Fiat” — and note that their analysis “runs parallel to recent discussions of determinism in Newtonian particle mechanics.”

A closer analogue, though only an analogue, is Doboszewski’s discussion [5, p. 12] of spacetimes that simply stop — Minkowski space with a spacelike slice and its future removed — for which he coins *Doomsday indeterminism*, remarking that “there is no physical reason for spacetime to come to an end abruptly in this way.” The comparison should be kept limited and it is not an equivalence: his truncations are produced by excising the future by hand, whereas here the finite endpoint is generated by the equation. That difference bears on his own desideratum, paraphrased from Earman and Norton, that “determinism may fail, but (one hopes) if it fails, it should fail because of reasons of physics, not because of a clever mathematical trick” [5, p. 2]. We claim no priority over that discussion and do not suggest the two constructions are of a kind.

Reading 3: termination as its own failure

A third position holds that the uniqueness-based formulation is the wrong test in the first place. Azhar and Namjoo [2] propose that indeterminism arises when future states are “(i) not unique . . . ; (ii) not specified; or (iii) ‘incoherent’”, and argue that a Laplacian criterion based on non-uniqueness alone “does not comprehensively capture the relevant sense of a lack of determination”. On such a view the branches here witness a failure of determination by ending, independently of how many of them there are.

What happens at $T + T_*$

The mathematical fact is bare: $x_T(t) \rightarrow +\infty$ as $t \uparrow T + T_*$, and no continuation past that time exists. What that *is* remains open, and at least three answers are available: (a) the branch is a history that terminates at a finite time; (b) the divergence signals that the model has broken down and is no longer describing anything; (c) $+\infty$ is not a state, so the solution leaves the state space and simply ceases to exist.

The mathematics is indifferent: Theorem 7 holds under all three, because it is a statement about a set of solutions and not about what solutions represent. The determinism verdict is not indifferent. The example bears on determinism only under (a), or under a criterion on which failure of continuation counts against determinism however the failure is described. **We therefore make no unconditional claim that any**

physical theory is indeterministic. Under (b) or (c) the example remains exactly what Theorem 7 says it is, and nothing more.

As to the positions, the example adds nothing. What it adds is a test case in which the three readings give visibly different verdicts about one state of one system, with no idealisation to argue over: the equation is scalar and autonomous, f is continuous everywhere and C^1 away from one point, and every quantity is exact. One point is common to all three readings: the distinction is not dissolved by demanding more regularity. The Lipschitz failure sits at the equilibrium alone, and the branches terminate because the equation is superlinear at infinity, which is ordinary behaviour.

8 What this note contributes

The phenomenon is classical and the distinction has substantial precedent, so the point of the example is not a theorem or a new criterion of determinism. It is that the whole thing is cheap: a scalar autonomous equation, elementary integrals, an exact blow-up time, an explicit family that can be drawn, and no general relativity, N -body mechanics or partial differential equations needed to check any step. Where the standard illustrations of the two failure modes are Norton’s dome [11] for non-uniqueness and the non-collision singularities of Xia [16] for finite-time escape, here both occur in one equation that fits on a line.

Limitations. Everything above concerns a scalar autonomous equation; the monotonicity argument of Step 1 uses the scalar order structure and does not survive to systems without new work. Nothing here is a theorem about Newtonian N -body dynamics. Appendix B gives a one-dimensional mechanical system exhibiting the mechanism, but it realises the mechanism, not this f . The interpretive question of Section 7 is left open, as it is in the literature cited there.

Conclusion. The example shows, in a simple setting, why a statement about uniqueness must specify the class of solutions being quantified over. Uniqueness among globally defined solutions and uniqueness among solutions that cannot be extended are different properties of the same law, and here they disagree at the same state, by the largest margin available.

A Two-sided time

The forward account says nothing about $x < 0$, and for this f , which is even, the negative half-line is populated: solutions can also arrive at the origin from below in finite time. Accounting for them identifies the solutions maximal in the standard sense and confirms that they are not the objects counted in Theorem 7.

Write $S_* = \int_{-\infty}^0 du/f(u)$. Two conditions are involved, and they are logically separate:

(H4a) $\int_{-1}^0 du/f < \infty$ — finite-time access to the equilibrium from below;

(H4b) $\int_{-\infty}^{-1} du/f < \infty$ — finite-time escape to $-\infty$ in backward time.

For an even f both follow from (H2) and (H3); in general neither follows from anything on the positive side.

Example 10 ((H4b) without (H4a)). Take $f(x) = x^{2/3} + x^2$ for $x \geq 0$ and $f(x) = |x| + x^2$ for $x < 0$. This is continuous, and the forward conditions hold because the positive half-line is unchanged. Since $1/(v+v^2) = 1/v - 1/(1+v)$ has antiderivative $\log(v/(1+v))$, we get $\int_1^\infty dv/(v+v^2) = \log 2$ but $\int_0^1 dv/(v+v^2) = \infty$. So (H4b) holds and (H4a) fails.

When T_* and S_* are both finite, the same classical mechanism applied to $\tilde{f}(y) = f(-y)$ in reversed time gives a two-parameter family: for $(S, T) \in [0, \infty]^2$ there is a solution whose zero set is $[-S, T]$, defined on $(-S - S_*, T + T_*)$, with the convention that an infinite parameter removes the corresponding bound. These are the maximal solutions in the standard sense, and $x \equiv 0$ is the only one defined on all of $\mathbb{R}$.

The point relevant to Section 5 falls out. For every S , the restriction to $[0, \infty)$ of the solution indexed by (S, T) is the same x_T — the zero set meets $[0, \infty)$ in $[0, T]$ whatever S is. So each forward-maximal solution extends backward in continuum-many ways, and none of them is maximal.

B A mechanical realisation

It is natural to ask whether a Newtonian particle can exhibit the mechanism. It can, though not in the way one first tries.

B.1 Not by steepening a dome

Norton's construction [11] places a particle on a frictionless surface of revolution, with r the arc length along a meridian from the apex and $h(r)$ the depth below it, giving $\ddot{r} = g h'(r)$. The obvious candidate here, $h(r) = \frac{1}{g}(\frac{2}{3}r^{3/2} + \frac{1}{3}r^3)$, does give $\ddot{r} = \sqrt{r} + r^2$, but it cannot be realised.

Proposition 11. *In a Norton-type model — frictionless, uniform gravity, arc-length meridional coordinate, purely meridional motion with zero angular momentum, and the particle remaining on the surface — no surface of revolution admits a solution reaching $r = +\infty$ in finite time.*

Proof. Parametrising the meridian by arc length forces $\rho'^2 + z'^2 = 1$ and hence $|h'| \leq 1$, so $|\ddot{r}| \leq g$. Integrating this bound twice from rest gives $r(t) \leq r(0) + \dot{r}(0)t + \frac{1}{2}gt^2$, which is finite for every finite t . $\square$

With $g = 9.8$ the candidate above already violates the constraint at $r \approx 2.85$. The bound genuinely uses the whole model: with angular momentum the equation gains a centrifugal term $(p_\theta^2/m^2) \rho'(r)/\rho(r)^3$, unbounded as $\rho \rightarrow 0$. So this closes one route and no more.

B.2 In a potential

Let $V(x) = -\left(\frac{2}{3}x^{3/2} + \frac{1}{3}x^3\right)$ for $x \geq 0$, so $\ddot{x} = -V'(x) = \sqrt{x} + x^2$, with $x(0) = \dot{x}(0) = 0$. Energy conservation from rest at the origin gives $\frac{1}{2}\dot{x}^2 = -V(x)$, that is $\dot{x} = G(x)$ with

$$G(x) = \left(\frac{4}{3}x^{3/2} + \frac{2}{3}x^3\right)^{1/2},$$

and conversely $GG' = \frac{d}{dx}\left(\frac{1}{2}G^2\right) = \sqrt{x} + x^2$, so solutions of the first-order equation solve the second-order one. The even extension $\hat{G}(x) = G(|x|)$ is continuous with $\hat{G}(0) = 0$ and $\hat{G} > 0$ off the origin; $G(x) \asymp x^{3/4}$ as $x \rightarrow 0^+$ and $\asymp x^{3/2}$ as $x \rightarrow \infty$, so (H2) and (H3) hold. Everything in Sections 3–4 therefore applies, with

$$T_*^{\text{mech}} = \int_0^\infty \frac{dx}{G(x)} = \frac{2}{3}\sqrt{\frac{3}{2}}2^{-1/3}B\left(\frac{1}{6}, \frac{1}{3}\right) \approx 5.4521363617.$$

The closed form is obtained as follows, and the reader can check every step. Factor $G^2 = \frac{4}{3}x^{3/2}\left(1 + \frac{1}{2}x^{3/2}\right)$ and substitute $s = \frac{1}{2}x^{3/2}$, so that $x = (2s)^{2/3}$ and $dx = \frac{2}{3}2^{2/3}s^{-1/3}ds$. Then

$$\frac{dx}{G(x)} = \frac{2}{3}\sqrt{\frac{3}{2}}2^{-1/3}s^{-5/6}(1+s)^{-1/2}ds,$$

and since $\int_0^\infty s^{a-1}(1+s)^{-a-b}ds = B(a, b)$ with $a = \frac{1}{6}$ and $b = \frac{1}{3}$, the integral is $\frac{2}{3}\sqrt{\frac{3}{2}}2^{-1/3}B\left(\frac{1}{6}, \frac{1}{3}\right)$ as stated.

This realises the *mechanism* of Theorem 7 in a Newtonian system — not the example of Section 2. The reduced right-hand side G is not $|x|^{2/3} + x^2$, and accordingly $T_*^{\text{mech}} \neq T_*$: a different member of the same class. Two caveats. The potential is unbounded below, the idealisation that attracts the standard objection to escape solutions [7]; and V is C^1 but not C^2 at the origin, the force being continuous and not locally Lipschitz there. Whether such a force law counts as a legitimate Newtonian system is the question the dome literature asks about $\sqrt{r}$ [9, 11]; we take no position.

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